

LaST-R1: Reinforcing Robotic Manipulation via Adaptive Physical Latent Reasoning

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Project page: <https://siriyeep.github.io/last-r1/>

Abstract

Robotic foundation models require reasoning over complex visual scenes to execute adaptive actions in dynamic environments. While recent studies on latent-reasoning Vision-Language-Action (VLA) models have demonstrated the capability to capture fine-grained physical dynamics, they remain predominantly confined to static imitation learning, severely limiting their adaptability and generalization. In this paper, we present **LaST-R1**, a novel reinforcement learning (RL) post-training framework designed to effectively harness “latent reasoning-before-acting” policies. Specifically, we propose **Latent-to-Action Policy Optimization (LAPO)**, a core RL algorithm that jointly optimizes the latent reasoning process and the action generation. By explicitly embedding latent Chain-of-Thought (CoT) reasoning directly within the RL optimization loop, LAPO stimulates profound physical world modeling, which in turn drives robust execution in interactive environments. Furthermore, an **adaptive latent CoT mechanism** is introduced, allowing the policy to dynamically modulate its reasoning horizon based on diverse environment states. Experiments show that LaST-R1 achieves a near-perfect 99.9% average success rate on the LIBERO benchmark with only one-shot supervised warm-up, significantly improving convergence speed and performance over prior state-of-the-art (SOTA) methods. In real-world deployments, LaST-R1 yields up to a 22.5% average improvement over SOTA supervised fine-tuning approach across four complex tasks, including both single-arm and dual-arm settings. Finally, LaST-R1 demonstrates strong generalization across simulated and real-world environments.

1 Introduction

Driven by large-scale pre-trained VLMs [1, 2, 3], Vision-Language-Action (VLA) models [4, 5, 6, 7, 8, 9, 10] have emerged as a promising paradigm for general robotic manipulation. Moving beyond direct observation-to-action mapping, recent architectures increasingly draw inspiration from Chain-of-Thought (CoT) reasoning [11]. While explicitly generating linguistic traces [12, 13, 14, 15] or future states [16, 17, 18, 19, 20] provides structured guidance before acting, it incurs non-negligible inference latency and discretization bottlenecks. This inherently restricts the model’s ability to capture continuous, high-frequency physical dynamics. Therefore, recent research [21, 22] has pivoted toward reasoning in a compact latent space that accommodates fine-grained, hard-to-verbalize physical knowledge, establishing a highly expressive reasoning foundation for complex manipulation.

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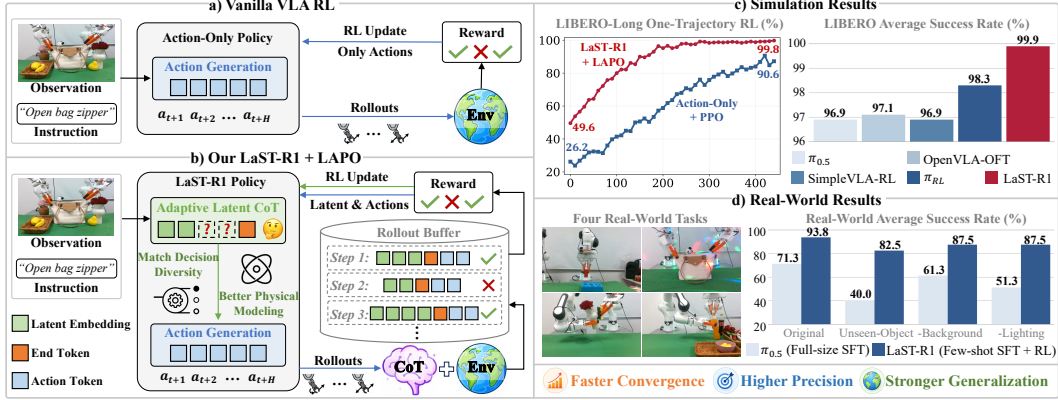


Figure 1: **The Overview of LaST-R1.** (a) Unlike vanilla RL baselines that strictly optimize actions, (b) our approach utilizes LAPO to jointly optimize an adaptive latent CoT alongside physical execution. By bridging cognitive reasoning and control, LaST-R1 achieves (c) faster convergence speed, higher success rate in simulation, and (d) stronger generalization capabilities in real-world.

Despite the advantages of latent reasoning, existing frameworks remain grounded in imitation learning, requiring massive, high-cost expert demonstrations. Their reliance on static datasets prevents closed-loop environmental interaction, leading to compounding errors and limited generalization. Concurrently, a nascent line of research [23, 24, 25, 26, 27, 28, 29, 30] has introduced online reinforcement learning (RL) for VLA post-training, improving exploration and robustness through trial-and-error interaction with the environment. However, as shown in Figure 1 (a), current methods are largely restricted to vanilla architectures that operate directly in the action space, bypassing the underlying physical reasoning process. Consequently, they lack interaction-driven learning to accurately model physical dynamics, limiting their ability to adapt to complex environments.

To this end, we propose **LaST-R1**, a novel RL post-training framework specifically developed to optimize “latent reasoning-before-acting” policies, as shown in Figure 1 (b). We first present the LaST-R1 VLA model, which tightly integrates latent CoT reasoning with action generation. Specifically, the model autoregressively produces latent reasoning embeddings to capture structured physical dynamics, serving as conditions for parallel action decoding. To ensure stable and expressive reasoning, these latent embeddings are grounded in global feature representations from a vision foundation model [31], providing a robust semantic and spatial prior. Building on this architecture, we propose **Latent-to-Action Policy Optimization (LAPO)**, a RL algorithm that jointly optimizes the latent reasoning process and action generation. Unlike prior RL approaches that operate solely within the action space [24, 23, 25], LAPO treats latent reasoning embeddings as implicit decision variables. By bridging this implicit deliberation with action generation through a unified step-level likelihood ratio, our method allows environmental rewards to directly shape the latent reasoning space, thereby concurrently optimizing the model’s physical comprehension and interactive robustness. Finally, an **adaptive latent CoT mechanism** is introduced to augment the framework, enabling the dynamic adjustment of the reasoning horizon based on task diversity. This allows the policy to allocate more reasoning steps for intricate manipulations, while maintaining fast execution for reactive responses, effectively balancing cognitive capacity and computational efficiency.

To empirically validate our framework, LaST-R1 VLA is first initialized with pre-trained Qwen3-VL-4B [2] and undergoes large-scale pre-training across diverse robotic manipulation datasets [32, 33, 34]. For specific downstream tasks, the policy is adapted through a supervised fine-tuning (SFT) warm-up before engaging in online RL post-training. During evaluation, our approach achieves a near-perfect 99.9% average success rate on the LIBERO benchmark with one-shot warmup, outperforming prior state-of-the-art (SOTA) baselines, with a particularly notable improvement of 5.3% on the LIBERO-Long suite (long-horizon tasks). In real-world deployments, LaST-R1 yields up to a 22.5% improvement over the SOTA SFT method across four complex tasks, reaching a 90% average success rate in both single-arm and dual-arm settings. Furthermore, systematic evaluations demonstrate that the proposed framework exhibits strong generalization in simulation environments, consistently outperforming vanilla action-only RL policies trained with standard PPO [35]. Notably, LaST-R1 achieves zero-shot generalization to unseen objects, backgrounds, and lighting conditions after LAPO post-training. In summary, our main contributions are as follows:

- 1) We present **LaST-R1**, a specialized RL post-training framework for “latent reasoning-before-acting” policies to enhance physical world modeling and robust action generation.
- 2) We propose **Latent-to-Action Policy Optimization (LAPO)**, a novel RL algorithm that jointly optimizes latent reasoning and action generation via a unified step-level likelihood ratio.
- 3) We design an **adaptive latent CoT mechanism** that dynamically adjusts the reasoning horizon based on task diversity, balancing reasoning capacity and computational efficiency.

2 Methodology

This section presents the proposed **LaST-R1** framework, as shown in Figure 2. We first review the problem formulation (Section 2.1), then introduce a unified VLA architecture with latent Chain-of-Thought (CoT) reasoning (Section 2.2). Next, we present **Latent-to-Action Policy Optimization (LAPO)**, which jointly optimizes latent reasoning and action generation (Section 2.3). Finally, we describe an **adaptive mechanism** that dynamically adjusts the reasoning horizon (Section 2.4).

2.1 Preliminaries

We formulate the Vision-Language-Action (VLA) model as a parameterized policy π_θ that maps multimodal observations (i.e., visual inputs and language instructions) to an action chunk $\mathbf{a}_{t:t+H}$ in $SE(3)$ space. For single-arm robots such as the Franka Research 3, the action is a 7-DoF end-effector control vector $\mathbf{a}_t \in \mathbb{R}^7$, comprising 3-DoF positional offsets, a 3-DoF orientation (represented as Euler angles), and a 1-DoF gripper state. For dual-arm systems, we extend this formulation to 14-DoF via vector concatenation. The policy can be optimized either through supervised fine-tuning (SFT) using expert demonstrations or via reinforcement learning (RL) via environment interaction.

SFT Formulation. In supervised fine-tuning, we have access to a dataset of expert demonstrations $\mathcal{D} = \{(\mathbf{s}_t, \mathbf{a}_{t:t+H})\}$, where $\mathbf{s}_t$ denotes the multimodal observations at timestep t . The policy is trained to imitate expert behavior by maximizing the conditional log-likelihood:

$$\mathcal{J}_{\text{SFT}}(\theta) = \mathbb{E}_{(\mathbf{s}_t, \mathbf{a}_{t:t+H}) \sim \mathcal{D}} [\log \pi_\theta(\mathbf{a}_{t:t+H} \mid \mathbf{s}_t)]. \quad (1)$$

While this objective enables effective behavioral cloning, the model’s capability and generalization remain inherently bounded by the quality, scale, and diversity of the training data.

RL Formulation. To surpass the limitations of static imitation, RL casts the sequential decision-making problem as an interactive process where the policy engages with the environment over trajectories of length $T + 1$. At each timestep t , the agent observes a state $\mathbf{s}_t$ and samples an action chunk $\mathbf{a}_{t:t+H} \sim \pi_\theta(\cdot \mid \mathbf{s}_t)$. The environment then transitions to the next state $\mathbf{s}_{t+1}$ and yields a scalar reward r_t . This interaction induces a trajectory τ , whose distribution is governed by the initial state and the environment dynamics. The objective is to maximize the expected discounted return:

$$\mathcal{J}_{\text{RL}}(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^T \gamma^t r_t \right], \quad \nabla_\theta \mathcal{J}_{\text{RL}}(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^T \nabla_\theta \log \pi_\theta(\mathbf{a}_{t:t+H} \mid \mathbf{s}_t) \hat{A}_t \right], \quad (2)$$

where $\gamma \in [0, 1)$ is the discount factor, and $\hat{A}_t$ denotes the advantage estimate, capturing the relative utility of the sampled action chunk $\mathbf{a}_{t:t+H}$ compared to the expected value at state $\mathbf{s}_t$.

2.2 LaST-R1 Model Architecture

As shown in Figure 2 (a), we propose a unified LaST-R1 VLA model that jointly models latent reasoning and action generation. We first present the overall pipeline, followed by a novel method for constructing compact latent representations of future dynamics.

Overall Pipeline. LaST-R1 VLA builds upon the Qwen3-VL-4B [2] architecture, integrating a SigLIP2-Large [36] visual encoder with an LLM backbone. Given an input image $I \in \mathbb{R}^{H \times W \times 3}$, the visual encoder first employs 2D-RoPE with interpolated absolute positional embeddings to transform spatial information into dense visual embeddings $f_v \in \mathbb{R}^{N_v \times 2560}$. Subsequently, these embeddings are concatenated with language embeddings $f_l \in \mathbb{R}^{N_l \times 2560}$ and processed by the LLM backbone, where N_v and N_l denote the respective sequence lengths. The model first generates latent reasoning embeddings, followed by action prediction. To fully leverage the strong reasoning capability of

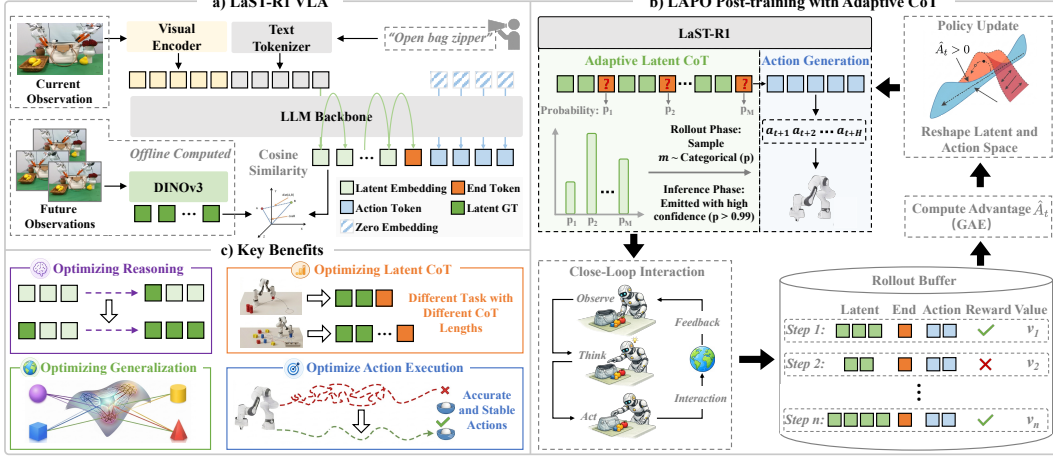


Figure 2: **The Framework of LaST-R1.** (a) LaST-R1 VLA is a unified model that takes visual observations and language instructions as input, where a vision foundation model provides physically grounded latent targets to guide latent CoT reasoning before action generation. (b) During LAPO RL post-training, LaST-R1 interacts with the environment in a closed loop manner, storing latents, actions, and rewards in a rollout buffer for jointly reshaping the latent and action spaces. It further enables adaptive reasoning by learning to emit the `<latent_end>` token based on predicted probabilities, dynamically adjusting the reasoning horizon across tasks. (c) Through LAPO, LaST-R1 achieves adaptive reasoning lengths across diverse tasks, improving generalization and execution stability.

the LLM, latent embeddings are generated in an autoregressive manner, with their representation detailed in the next paragraph. To support action generation, we normalize and discretize continuous actions into tokens, extending the LLM vocabulary accordingly [4]. We adopt a parameter-free action tokenizer to map between continuous actions and discrete tokens, while employing parallel decoding to improve inference efficiency [5]. To enable RL post-training, we introduce a value head composed of a 4-layer MLP to estimate state values, which shares the same backbone with the VLA [28, 23].

Latent Representation. To model temporal environmental dynamics, LaST-R1 VLA autoregressively infers future states over a horizon N_z , yielding N_z latent reasoning embeddings. Prior methods extract these latents via average pooling [21] or auxiliary learnable parameters [22], which often sacrifices fine-grained spatial structures or injects undesired inductive biases into the learned latent space. To overcome this, we construct latent embeddings using DINOv3 [31], a state-of-the-art vision foundation model. Specifically, we extract its `<CLS>` token ($f_d \in \mathbb{R}^{1 \times 4096}$) as a holistic image embedding and apply top- k ($k = 2560$) selection based on feature magnitude along the channel dimension. These selected latent targets align with the hidden size of the VLA embedding while preserving the most salient visual components. By harnessing DINOv3’s structurally rich and semantically dense feature space, the extracted `<CLS>` token serves as a strong semantic prior, facilitating accurate modeling of physical dynamics. Crucially, these latent targets are precomputed offline, effectively decoupling the foundation model from the policy to ensure no additional overhead during both training and inference. We systematically compare this approach with prior latent representation formats in Section 3.2 to further validate its effectiveness.

2.3 Latent-to-Action Policy Optimization

To effectively train the proposed *latent reasoning-before-acting* policy LaST-R1 VLA, we introduce **Latent-to-Action Policy Optimization (LAPO)**, a RL framework that jointly optimizes latent reasoning and action generation, as shown in Figure 2 (b). In our formulation, the policy first generates latent embeddings, which condition subsequent action prediction. While prior works [28, 37] have established PPO [35] as an effective post-training method for VLA models, they operate solely in the action space and overlook intermediate reasoning processes that are integral to decision-making. To address this limitation, we propose LAPO, which treats latent embeddings as *implicit decision variables* and introduces a unified optimization framework that enables environment reward signals to shape both reasoning and action generation spaces across the entire trajectory.

Rollout Collection. At each environment step t , given the current state $\mathbf{s}_t$ comprising visual observations and language instructions, the policy first generates a sequence of latent embeddings $\mathbf{Z}_t^{\text{old}} = \{\mathbf{z}_{t,i}^{\text{old}}\}_{i=1}^{N_z}$ autoregressively: $\mathbf{z}_{t,i}^{\text{old}} \sim \pi_\theta(\cdot \mid \mathbf{s}_t, \mathbf{z}_{t,<i}^{\text{old}})$. Conditioned on these latents, a sequence of discrete action tokens $\mathbf{C}_t = \{\mathbf{a}_{t,j}\}_{j=1}^{N_a}$ is then produced, where N_a denotes the total number of tokens representing the H -step action chunk. Specifically, we reuse the KV cache from the latent generation phase and decode the action tokens in parallel via a single forward pass with bidirectional attention over N_a placeholder vectors. We further insert a special `<latent_end>` token between the latent and action sequences to signify the completion of reasoning. The final hidden embedding of this token is routed into a value head to output the state value estimation v_t . To formulate a unified action distribution for decision step t , we compute the joint log-probability of the action sequence by summing over the individual tokens: $\log \pi_\theta(\mathbf{C}_t \mid \cdot) = \sum_{j=1}^{N_a} \log \pi_\theta(\mathbf{a}_{t,j} \mid \cdot)$. During rollout, we store the sampled latent sequences $\mathbf{Z}_t^{\text{old}}$, action token sequences $\mathbf{C}_t$, their corresponding log-probabilities, and the estimated state values. Ultimately, a scalar reward is assigned based on task success or failure, yielding complete trajectories that are subsequently used to compute the advantage estimates $\hat{A}_t$ via Generalized Advantage Estimation (GAE) for policy optimization.

Policy Optimization. During policy updates, the model processes the identical rollout context to yield updated continuous latent embeddings $\mathbf{Z}_t^\theta = \{\mathbf{z}_{t,i}^\theta\}_{i=1}^{N_z}$ and updated discrete action log-probabilities. Because the advantage $\hat{A}_t$ is computed at the decision-step level, we define a step-level ratio $r_t(\theta)$ for both the action sequence and the continuous latents. For the discrete action sequence, the ratio is computed via the joint probabilities: $r_t^a(\theta) = \exp(\log \pi_\theta(\mathbf{C}_t \mid \cdot) - \log \pi_{\theta_{\text{old}}}(\mathbf{C}_t \mid \cdot))$. For the continuous latent embeddings, we approximate the distribution of rollout latents $\mathbf{Z}_t^{\text{old}}$ using an isotropic Gaussian centered at the current policy output $\mathbf{Z}_t^\theta$. By summing the squared Euclidean distances over the entire latent sequence length N_z , we define the step-level latent ratio surrogate as:

$$r_t^z(\theta) = \frac{\pi_\theta(\mathbf{Z}_t^{\text{old}} \mid \cdot)}{\pi_{\theta_{\text{old}}}(\mathbf{Z}_t^{\text{old}} \mid \cdot)} = \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{N_z} \|\mathbf{z}_{t,i}^{\text{old}} - \mathbf{z}_{t,i}^\theta\|^2\right), \quad (3)$$

where σ is a fixed hyperparameter regulating the latent variance. For a more intuitive, step-by-step derivation of $r_t^z(\theta)$, please refer to Appendix B. By jointly optimizing reasoning latents and actions, we formulate the joint LAPO clipped surrogate loss as an expectation over all valid decision steps t :

$$\mathcal{L}_{\text{policy}}(\theta) = -\mathbb{E}_t \left[\sum_{k \in \{z, a\}} \min\left(r_t^k(\theta) \hat{A}_t, \text{clip}(r_t^k(\theta), 1 - \epsilon_{\min}, 1 + \epsilon_{\max}) \hat{A}_t\right) \right], \quad (4)$$

where $\epsilon_{\min}$ and $\epsilon_{\max}$ are the clipping thresholds. This objective enables the environment reward to shape both the action space and the internal reasoning space simultaneously. Specifically, when $\hat{A}_t > 0$, the optimization explicitly minimizes the latent distance $\sum_{i=1}^{N_z} \|\mathbf{z}_{t,i}^{\text{old}} - \mathbf{z}_{t,i}^\theta\|^2$, effectively pulling the current policy’s latent representations toward the “good-reasoning” manifolds that facilitated successful trajectories. In practice, to achieve better optimization flexibility, we decouple the joint policy objective $\mathcal{L}_{\text{policy}}(\theta)$ into two separate components: the action loss $\mathcal{L}_{\text{action}}(\theta)$ (for $k = a$) and the latent loss $\mathcal{L}_{\text{latent}}(\theta)$ (for $k = z$). Finally, the total training loss unifies these policy objectives with state-value estimation, governed by weighting coefficients λ_1 and λ_2 :

$$\mathcal{L}_{\text{total}}(\theta) = \mathcal{L}_{\text{action}}(\theta) + \lambda_1 \mathcal{L}_{\text{latent}}(\theta) + \lambda_2 \mathcal{L}_{\text{value}}(\theta), \quad (5)$$

where $\mathcal{L}_{\text{value}}(\theta) = \mathbb{E}_t \left[(v_t - \hat{R}_t)^2 \right]$ is the mean squared error (MSE) against the target return $\hat{R}_t$.

2.4 Adaptive Latent CoT Reasoning

While fixed-length reasoning provides a baseline, it imposes a static cognitive horizon that fails to capture the inherent diversity of robotic tasks. Enforcing a fixed latent length N_z incurs unnecessary computational overhead on highly predictable motions and restricts the reasoning capacity required for maneuvers demanding intricate cognitive planning. To address this, we introduce an **adaptive latent CoT mechanism** optimized via our proposed RL framework. By allowing the model to adapt its reasoning length based on reward signals, we enable an “early-exit” strategy that dynamically tailors the latent reasoning horizon to the specific demands of each task.

Dynamic Latent Generation. To enable adaptive reasoning, we shift the `<latent_end>` token from a deterministic sequence terminator to a dynamically emitted transition signal within the latent generation process. During autoregressive generation, if the policy predicts the `<latent_end>` token with a sufficiently high confidence probability $p \geq 0.99$ (where $p \in [0, 1]$), indicating that the model is highly certain the reasoning phase should conclude at the current decision step, the latent generation terminates and the model transitions to action prediction. To stabilize the training process and prevent erratic fluctuations in latent reasoning length, we restrict the emission of the `<latent_end>` token to a predefined set of M candidate positions, bounded by a maximum reasoning length $N_{\max}$.

Exploration via Length Sampling. During RL rollouts, exploring various reasoning lengths is crucial for discovering the optimal cognitive horizon across diverse environmental states. We treat the reasoning length as a stochastic decision to encourage environmental interaction. Specifically, we extract the pre-softmax logits l associated with the `<latent_end>` token across all M predefined candidate positions, and then apply a temperature parameter β to control the exploration entropy, yielding a normalized categorical distribution over the candidate lengths:

$$p_m = \frac{\exp(l_m/\beta)}{\sum_{i=1}^M \exp(l_i/\beta)}, \quad \forall m \in \{1, \dots, M\}. \quad (6)$$

The reasoning length index $m \sim \text{Categorical}(p_1, \dots, p_M)$ is sampled to dictate the number of generated latent embeddings, facilitating exploration during rollout to discover adaptive cognitive horizons. During inference, the model adopts a confidence-based exit strategy for efficient execution.

Adaptive Length Optimization. To optimize the adaptive length selection, the decision of *when* to stop reasoning must be explicitly supervised. We introduce an additional policy loss term for the `<latent_end>` token. Let $\mathbf{z}_t^{\text{end}}$ denote the transition token emitted at the sampled position m . We record its log-probability $\log \pi_\theta(\mathbf{z}_t^{\text{end}} | \cdot)$ during rollout. During the policy update, we construct a discrete likelihood ratio for the transition token: $r_t^{\text{end}}(\theta) = \exp(\log \pi_\theta(\mathbf{z}_t^{\text{end}} | \cdot) - \log \pi_{\theta_{\text{old}}}(\mathbf{z}_t^{\text{end}} | \cdot))$. For models equipped with adaptive reasoning, the total training objective is augmented to include this transition-specific term $\mathcal{L}_{\text{end}}(\theta)$, weighted by a coefficient λ_3 :

$$\mathcal{L}_{\text{total}}(\theta) = \mathcal{L}_{\text{action}}(\theta) + \lambda_1 \mathcal{L}_{\text{latent}}(\theta) + \lambda_2 \mathcal{L}_{\text{value}}(\theta) + \lambda_3 \mathcal{L}_{\text{end}}(\theta). \quad (7)$$

Guided by $\hat{A}_t$, this objective optimizes reasoning efficiency by rewarding shorter paths in predictable states while penalizing insufficient reasoning in scenarios requiring more cognitive planning.

3 Experiments

Following prior works [8, 9], LaST-R1 is first pre-trained on a custom-designed large-scale dataset, detailed in Appendix C.1. We then systematically evaluate our framework: Section 3.1 compares our method against SOTA baselines on the LIBERO benchmark, and Section 3.2 provides ablation studies on our core designs. Furthermore, Section 3.3 validates the performance of LaST-R1 in real-world robotic deployments, while Section 3.4 thoroughly investigates its generalization capabilities.

3.1 Simulation Experiment

Settings. We validate our method on four LIBERO task suites [38]: LIBERO-Spatial, LIBERO-Object, LIBERO-Goal, and LIBERO-Long, each comprising 10 distinct tasks. All setups feature a simulated Franka Panda with a single front-view observation at 256×256 resolution. Following [24], we adopt a two-stage training pipeline. We first perform an SFT warm-up using a single randomly selected data per task, followed by online RL to iteratively collect rollouts and update the policy within the simulator. More training details are placed in Appendix D.2 and D.3. Finally, performance is measured by the average Success Rate (SR) across 50 held-out test scenarios for each task.

Baselines. We conduct a comprehensive comparison against both SFT-only and RL-trained models. All SFT baselines are trained on the full training dataset (50 trajectories per task). Among the RL-trained baselines, GRAPE [39] utilizes DPO [40], VLA-RL [25] and π_{RL} [23] rely on PPO [35], TGRPO [41], RLinf-GRPO [37], and SimpleVLA-RL [24] are optimized via GRPO [42]. All RL-trained baselines follow their official warm-up strategies, as detailed in Table 1.

Result Analysis. Table 1 presents the quantitative comparison across the LIBERO benchmark. Our proposed LaST-R1 achieves state-of-the-art performance, recording a near-perfect average

Table 1: **Comparison on the LIBERO.** For RL, we use single-trajectory warm-up and single-view data. Specifically, $\dagger$ denotes full-trajectory warm-up, and $\ddagger$ indicates two-camera views training.

Models	Paradigm	Spatial		Object		Goal		Long		Average	
		SR $\uparrow$	Rank $\downarrow$	SR $\uparrow$	Rank $\downarrow$	SR $\uparrow$	Rank $\downarrow$	SR $\uparrow$	Rank $\downarrow$	SR $\uparrow$	Rank $\downarrow$
OpenVLA [4]	SFT	84.7	12	88.4	12	79.2	12	53.7	12	76.5	12
GR00T-N1 [43]	SFT	94.4	8	97.6	8	93.0	8	90.6	7	93.9	8
π_0 [6]	SFT	96.8	7	98.8	4	95.8	7	85.2	8	94.2	7
$\pi_{0.5}$ [7]	SFT	98.8	4	98.2	7	98.0	5	92.4	5	96.9	5
OpenVLA-OFT [5]	SFT	97.6	6	98.4	6	97.9	6	94.5	2	97.1	4
GRAPE $\dagger$ [39]	RL	88.5	11	92.1	10	83.1	9	57.2	11	80.2	11
TGRPO $\dagger$ [41]	RL	90.4	9	92.2	9	81.0	11	59.2	10	80.7	10
VLA-RL $\dagger$ [25]	RL	90.2	10	91.8	11	82.2	10	59.8	9	81.0	9
SimpleVLA-RL [24]	RL	98.2	5	98.7	5	98.8	3	91.7	6	96.9	5
RLinf-GRPO [37]	RL	98.9	3	99.7	3	98.3	4	93.6	4	97.6	3
π_{RL} $\ddagger$ [23]	RL	99.6	2	100.0	1	99.6	2	94.0	3	98.3	2
LaST-R1 (Ours)	RL	99.8	1	100.0	1	100.0	1	99.8	1	99.9	1

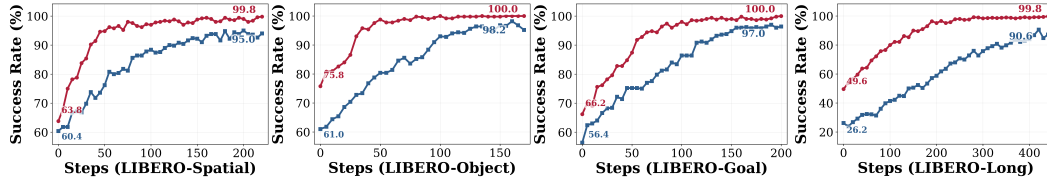


Figure 3: **One-shot Online RL Learning Curves on LIBERO.** We compare our proposed LaST-R1 optimized via LAPO (red), against the standard Action-Only baseline optimized via PPO (blue).

success rate of 99.9% and ranking first across all four task suites. Notably, despite using only a single trajectory for warm-up, our method outperforms strong SFT baselines such as $\pi_{0.5}$ (96.9%) and OpenVLA-OFT (97.1%), which heavily rely on complete expert datasets. When compared to the leading RL-based method π_{RL} under the identical one-trajectory setting, LaST-R1 maintains a consistent advantage. This gap becomes particularly pronounced on the highly challenging LIBERO-Long suite (99.8% vs. 94.0%), demonstrating that our framework possesses superior capabilities in handling complex and long-horizon manipulation tasks. Ultimately, these results demonstrate that LaST-R1 substantially improves the physical world modeling capability of VLA models, leading to near-perfect action performance even with one-shot training data. Meanwhile, Figure 3 presents the learning curves across four task suits in LIBERO. LaST-R1 trained with LAPO achieves significantly faster convergence and higher success rates compared to the Action-Only baseline optimized via PPO. The baseline policy struggles with optimization efficiency, especially in the LIBERO-Long task suite, requiring significantly more steps yet still converging to sub-optimal performance. The results show that introducing latent CoT optimization during post-training acts as an effective cognitive buffer, smoothing the RL optimization landscape and improving sample efficiency. Furthermore, additional analyses are provided in Appendix E.2 and E.3, showing how this optimized reasoning mechanism yields both adaptive reasoning lengths across tasks and optimized execution episode lengths.

3.2 Ablation Studies

To validate our core designs, we conduct ablation studies evaluated on the LIBERO-Spatial suite unless otherwise specified. **(1) Effectiveness of Latent Reasoning in SFT and RL Post-Training.** As shown in Figure 3, one-shot SFT with latent reasoning (LaST-R1 Warm-up) achieves a 63.9% average SR, surpassing the Action-Only baseline (51.0%). The improvement is particularly pronounced on LIBERO-Long (long-horizon tasks), where incorporating latent CoT boosts performance from 26.2% to 49.6%. After RL post-training, our full method (LaST-R1 + LAPO) further amplifies this advantage, achieving 99.9% SR across all task suites and significantly outperforming Action-Only + PPO (95.2%). These results demonstrate that our latent reasoning mechanism enhances physical understanding, leading to more effective action optimization, as further evidenced by action-to-vision attention (Appendix F.1). **(2) Design Choices for Latent Representation.** As shown in Figure 4 (a), we study the impact of different latent representation construction methods by comparing our DINOv3-based approach with three alternatives: averaging VLA visual encoder features [44] via global pooling [21], downsampling them with convolution [22], and extracting latents using a Q-Former [45]. Detailed implementations of these variants are deferred to Appendix D.1. Evaluated

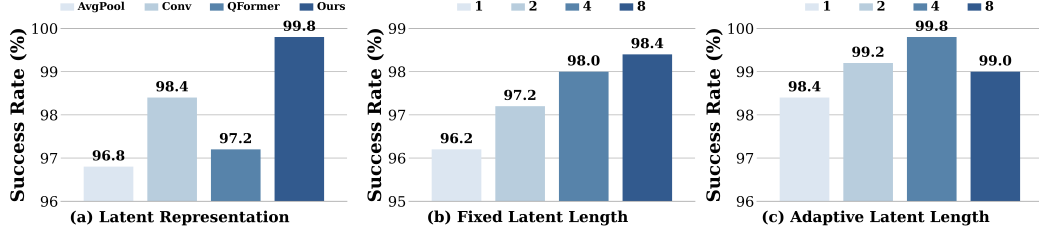


Figure 4: **Ablation Studies.** We evaluate (a) latent representation methods, (b) different fixed latent CoT lengths, and (c) adaptive CoT length with varying `<latent_end>` emission strategy.

after RL post-training, our DINOv3 representation achieves the highest SR of 99.8%, outperforming Convolution (98.4%), Q-Former (97.2%), and Global Pooling (96.8%). This validates that leveraging pre-trained global representations from DINOv3 better preserves intricate spatial and semantic structures, while the top- k selection scheme maximally retains dynamic information. **(3) Impact of Latent Reasoning Length.** We ablate the CoT horizon by varying the fixed latent sequence length $N_z \in \{1, 2, 4, 8\}$, as shown in Figure 4 (b). Compared to the Action-Only baseline (no latent reasoning, 95.0% SR), performance monotonically increases from 96.2% (1 token) to 98.4% (8 tokens). This indicates that a longer cognitive horizon effectively encodes richer condition information for downstream actions. Moreover, we observe that the performance gain from 4 to 8 tokens is marginal. To strike an optimal balance between task accuracy and inference speed, we cap the maximum reasoning sequence at 8 tokens. **(4) Adaptive CoT Length and `<latent_end>` Emission Strategy.** Finally, we validate our adaptive latent CoT mechanism in Figure 4 (c). Given a maximum reasoning length $N_{\max} = 8$, we ablate the number of valid candidate positions M for emitting `<latent_end>`. Specifically, we evaluate $M \in \{1, 2, 4, 8\}$ by uniformly distributing the candidate positions across the latent sequence. For example, $M = 1$ allows emission only after the 8th latent token, whereas $M = 8$ permits emission after every latent token. We find that $M = 4$ achieves the best performance (99.8%). While adaptive termination generally improves both efficiency and performance over the fixed-length baseline (98.4%), setting M too large ($M = 8$, 99.0%) slightly degrades performance. Therefore, a moderately constrained adaptive strategy ($M = 4$) offers the best trade-off between reasoning flexibility and optimization stability. More ablation studies are provided in Appendix E.1.

3.3 Real-World Experiment

Settings. We validate our method on four practical real-world manipulation tasks using Franka Research 3 arms, including one single-arm task and three dual-arm tasks: (1) *Insert hexagon block*, (2) *Open bag zipper*, (3) *Wipe vase with sponge*, and (4) *Open bottle cap*. For each task, we collect expert demonstrations using a space mouse. Given the significant domain gap between simulation and real-world deployment, we follow recent real-world VLA RL methods [26, 27] and adopt our proposed *latent reasoning-before-acting* RL framework. To improve RL update efficiency, we incorporate Low-Rank Adaptation (LoRA) [46] into all attention layers of LaST-R1, updating only the LoRA parameters. Additional implementation details are provided in Appendix D.4. For observations, we use one third-person camera and two wrist cameras, with input resolutions matching those in simulation. All policies are evaluated over 20 rollouts under varied positions, with task success determined by human evaluation. To highlight the advantages of our RL post-training framework, we compare LaST-R1 with the SOTA VLA model $\pi_{0.5}$, trained via full SFT on 100 expert trajectories.

Result Analysis. The “Original” column in Table 2 demonstrates the remarkable data efficiency of LaST-R1, where an initial warm-up with only 30 trajectories followed by online RL significantly outperforms the $\pi_{0.5}$ baseline trained with 100 expert trajectories via full supervised fine-tuning. Overall, this paradigm boosts the average SR substantially from an initial 52.5% to a final 93.75% ($\pm 1.25\%$ std), surpassing the 71.25% ($\pm 4.5\%$ std) average SR of the baseline $\pi_{0.5}$. The standard deviation (std) is computed over three independent runs. Specifically, the proposed RL method enhances visually guided precision in single-arm tasks by doubling the success rate of *Insert hexagon block* from 45% to 90%, while also improving dual-arm coordination for contact-rich manipulation like the *Open bag zipper* and *Wipe vase with sponge* tasks. Furthermore, the dramatic performance surge from 45% to 95% in the *Open bottle cap* task indicates that our post-training fundamentally strengthens the policy’s ability to model temporally evolving physical interactions. More visualizations and real-world execution videos are provided in Appendix F.2 and the supplementary material.

Table 2: **Comparison on Real-World Tasks.** Evaluation includes the original training scenario and three generalization settings: unseen objects, background variations, and lighting conditions. Example cases are shown below, with changed configurations highlighted in **red boxes**.

Methods	Insert hexagon block				Open bag zipper			
	Original	Unseen-Object	-Background	-Lighting	Original	Unseen-Object	-Background	-Lighting
$\pi_{0.5}$ [7] (Full-size SFT)	65	35 (-30%)	55 (-10%)	40 (-25%)	75	30 (-45%)	70 (-5%)	60 (-15%)
LaST-R1 (Few-shot SFT $\rightarrow$ RL)	45 $\rightarrow$ 90	75 (-15%)	85 (-5%)	80 (-10%)	55 $\rightarrow$ 95	80 (-15%)	95 (-0%)	90 (-5%)

Methods	Wipe vase with sponge				Open bottle cap			
	Original	Unseen-Object	-Background	-Lighting	Original	Unseen-Object	-Background	-Lighting
$\pi_{0.5}$ [7] (Full-size SFT)	75	45 (-30%)	65 (-10%)	50 (-25%)	70	50 (-20%)	55 (-15%)	55 (-15%)
LaST-R1 (Few-shot SFT $\rightarrow$ RL)	65 $\rightarrow$ 95	80 (-15%)	90 (-5%)	95 (-0%)	45 $\rightarrow$ 95	95 (-0%)	80 (-15%)	85 (-10%)

Insert hexagon block & Generalization scenario examples

Open bag zipper (Execution process)

Wipe vase with sponge (Execution process)

Open bottle cap (Execution process)

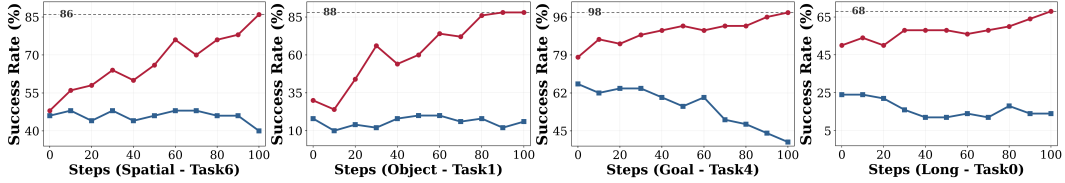


Figure 5: **Generalization Analysis on LIBERO.** While the OOD performance of the Action-Only PPO baseline (blue) stagnates, our LaST-R1 with LAPO (red) demonstrates continuous improvement.

3.4 Generalization Analysis

We evaluated LaST-R1’s generalization in both simulation and the real world. **(1) Simulation.** Following the setup of prior work [24], for each task suite in LIBERO, we perform online RL on 9 “seen” tasks and hold out 1 task for out-of-distribution (OOD) evaluation, matching our one-trajectory warm-up setting. As shown in Figure 5, the standard Action-Only + PPO baseline suffers from severe overfitting, its OOD performance stagnates and even degrades on the LIBERO-Goal and LIBERO-Long suites. Conversely, our LaST-R1 + LAPO policy demonstrates continuous, significant OOD improvements across all suites. This demonstrates that our latent reasoning with LAPO progressively improve the model’s understanding of scene and physical dynamics via environmental interaction, enabling it to extract transferable spatial and semantic concepts rather than overfitting to the training distribution. Additional experiments are presented in Appendix E.4. **(2) Real-World.** This superior adaptability seamlessly transfers to physical deployments. As shown in Table 2, when exposed to severe unseen perturbations, the full-size SFT baseline $\pi_{0.5}$ suffers catastrophic performance degradation, evidenced by up to a 45% drop when interacting with unseen objects. In stark contrast, LaST-R1 (Few-shot SFT $\rightarrow$ RL) maintains highly resilient execution. It successfully confines performance drops to within 15% for novel objects and demonstrates near-zero degradation against diverse background and lighting variations. This confirms that optimizing latent reasoning through environmental feedback intrinsically fortifies LaST-R1 VLA against environmental uncertainties.

4 Conclusion

In this work, we present LaST-R1, a tailored RL post-training framework for “latent reasoning-before-acting” policies with a unified VLA model coupling latent reasoning and action generation. By introducing latent Chain-of-Thought reasoning grounded in global visual representations, our approach enables structured modeling of physical dynamics prior to action execution. Building on this design, we propose Latent-to-Action Policy Optimization (LAPO), a novel RL paradigm that jointly optimizes reasoning and action spaces, allowing environmental feedback to shape both external behavior and internal reasoning processes. Furthermore, we introduce an adaptive latent CoT mechanism that dynamically adjusts the reasoning horizon based on task diversity, improving both

efficiency and performance. Experimental results highlight the importance of integrating physical latent reasoning into VLA learning and show that jointly optimizing reasoning and action through RL is a promising direction for improving generalization and physical understanding. We hope this work provides a foundation for scaling latent reasoning-driven policies in complex robotic manipulation.

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A Related Work

Vision-Language-Action (VLA) Models. Based on pretrained vision-language models (VLMs), VLA models [8, 6, 10, 47, 48] have demonstrated promising performance in robotic control with discrete action generation [4, 49] or continuous action representation. Recent frameworks have diversified into regression-based [5], diffusion-based [50, 9, 17, 47], and flow-matching-based architectures [43, 7, 51]. To further enhance precision and spatial awareness, several studies incorporate 3D spatial information or point-cloud reasoning into the VLA backbone [52, 53], significantly boosting performance in complex manipulation tasks. Inspired by Chain-of-Thought (CoT) reasoning in VLMs [11], recent research focuses on endowing VLAs with powerful reasoning capabilities. This includes generating intermediate text [7, 14], visual [22, 16], or multimodal plans [17], notably through methods like Embedded CoT [15] to strengthen spatial reasoning. However, these explicit decoding processes inevitably introduce inference latency. To alleviate this, further research explores introducing latent reasoning [54, 55] into the action generation process [56, 21, 22], balancing performance gains with execution efficiency. While these VLA frameworks have evolved significantly, the field remains predominantly confined to the imitation learning paradigm, where the lack of environmental interaction hinders the models’ ability to generalize beyond static datasets or further refine their internal reasoning for robust control.

Reinforcement Learning (RL) for VLA Models. To overcome the limitations of static imitation learning, Reinforcement Learning (RL) has been increasingly adopted to fine-tune VLA models through environmental feedback [57]. Early efforts focus on offline preference alignment [58, 39], such as GRAPE [39], which employs Direct Preference Optimization (DPO) [40] to align robot trajectories with human intent. To further enhance closed-loop robustness, online RL frameworks have emerged [28, 24, 23, 25, 27, 59, 26, 37]. For example, VLA-RL [25] and RL4VLA [28] utilize Proximal Policy Optimization (PPO) [35] to improve generalization in unseen environments, while SimpleVLA-RL [24] and TGRPO [41] apply Group Relative Policy Optimization (GRPO) [42] to achieve efficient post-training without complex reward modeling. Depending on the action space, these methods optimize either discrete token probabilities [24, 25] or continuous action sequences [23]. However, a critical gap remains: current RL-based VLA methods predominantly support vanilla architectures, focusing exclusively on action-space supervision. They fail to address how to conduct RL post-training for reasoning-based VLAs, leaving the intrinsic link between the internal "thought" process and the final "act" unoptimized. Therefore, we present LaST-R1, a RL post-training framework with a unified VLA model that sequentially performs latent reasoning and action generation. Building on this, we propose Latent-to-Action Policy Optimization (LAPO) to concurrently optimize both the reasoning trajectory and the action generation, achieving superior robustness and precision in complex manipulation tasks.

B Derivation of the Latent Ratio

In our Latent-to-Action Policy Optimization (LAPO) framework, we optimize both the action tokens and the continuous latent reasoning embeddings. While the likelihood ratio for action tokens is straightforward to compute via categorical probabilities, the ratio for the continuous latent embeddings requires defining a density function over the high-dimensional continuous space to estimate. Here, we provide the detailed mathematical derivation for the decision step-level latent ratio $r_t^z(\theta)$.

1. Distribution Assumption

Let D denote the dimensionality of a single latent reasoning embedding. During policy optimization, we approximate the distribution of the generated continuous latents using an isotropic Gaussian distribution [60] centered at the current policy’s output $\mathbf{z}_{t,i}^\theta$. We assume a fixed, predefined variance σ^2 across all dimensions. Thus, the probability density function (PDF) for a latent variable $\mathbf{x}$ generated by the current policy θ for the i -th latent embedding is given by:

$$\pi_\theta(\mathbf{x}) = \frac{1}{(2\pi\sigma^2)^{D/2}} \exp\left(-\frac{1}{2\sigma^2}\|\mathbf{x} - \mathbf{z}_{t,i}^\theta\|^2\right). \quad (8)$$

2. Evaluating the Numerator $\pi_\theta(\mathbf{z}_{t,i}^{\text{old}} \mid \cdot)$ and Denominator $\pi_{\theta_{\text{old}}}(\mathbf{z}_{t,i}^{\text{old}} \mid \cdot)$

To compute the ratio for a single latent embedding sampled during a previous rollout, we must evaluate both the new policy θ and the old policy θ_{old} on the collected old latent embedding $\mathbf{z}_{t,i}^{\text{old}}$.

For the **numerator** $\pi_{\theta}(\mathbf{z}_{t,i}^{\text{old}} \mid \cdot)$ (the probability of the old latent embedding under the *new* policy), we substitute $\mathbf{x} = \mathbf{z}_{t,i}^{\text{old}}$ into the PDF:

$$\pi_{\theta}(\mathbf{z}_{t,i}^{\text{old}} \mid \cdot) = \frac{1}{(2\pi\sigma^2)^{D/2}} \exp\left(-\frac{1}{2\sigma^2}\|\mathbf{z}_{t,i}^{\text{old}} - \mathbf{z}_{t,i}^{\theta}\|^2\right). \quad (9)$$

For the **denominator** $\pi_{\theta_{\text{old}}}(\mathbf{z}_{t,i}^{\text{old}} \mid \cdot)$ (the probability of the old latent embedding under the *old* policy), the mean of the old policy θ_{old} is, by definition, the sampled latent embedding $\mathbf{z}_{t,i}^{\text{old}}$. Consequently, the squared Euclidean distance becomes zero, i.e., $\|\mathbf{z}_{t,i}^{\text{old}} - \mathbf{z}_{t,i}^{\text{old}}\|^2 = 0$. The exponential term simplifies to $\exp(0) = 1$, leaving only the normalization constant:

$$\pi_{\theta_{\text{old}}}(\mathbf{z}_{t,i}^{\text{old}} \mid \cdot) = \frac{1}{(2\pi\sigma^2)^{D/2}}. \quad (10)$$

3. Computing the Single Latent Embedding Ratio

By dividing the numerator $\pi_{\theta}(\mathbf{z}_{t,i}^{\text{old}} \mid \cdot)$ by the denominator $\pi_{\theta_{\text{old}}}(\mathbf{z}_{t,i}^{\text{old}} \mid \cdot)$, the complex normalization constant $\frac{1}{(2\pi\sigma^2)^{D/2}}$ perfectly cancels out. The ratio for a single latent embedding i is elegantly reduced to the negative exponentiated distance:

$$r_{t,i}^z(\theta) = \frac{\pi_{\theta}(\mathbf{z}_{t,i}^{\text{old}} \mid \cdot)}{\pi_{\theta_{\text{old}}}(\mathbf{z}_{t,i}^{\text{old}} \mid \cdot)} = \exp\left(-\frac{1}{2\sigma^2}\|\mathbf{z}_{t,i}^{\text{old}} - \mathbf{z}_{t,i}^{\theta}\|^2\right). \quad (11)$$

4. Sequence-Level Latent Reasoning Embeddings Aggregation

The latent reasoning process consists of a sequence of N_z latent embeddings. Assuming the continuous latents are conditionally independent given the rollout context, the joint ratio for the entire latent sequence $\mathbf{Z}_t$ is the product of the individual embedding ratios. Since the product of exponentials equals the exponential of their sum, we arrive at our final formulation:

$$r_t^z(\theta) = \prod_{i=1}^{N_z} r_{t,i}^z(\theta) = \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{N_z} \|\mathbf{z}_{t,i}^{\text{old}} - \mathbf{z}_{t,i}^{\theta}\|^2\right). \quad (12)$$

C Dataset Details

C.1 Large Scale Pre-Training Datasets

To ensure LaST-R1 inherits a robust foundation of motor primitives and physical common sense, we curated a diverse corpus of 400K trajectories (28M frames) from the Open-X-Embodiment [32], DROID [34], and RoboMIND [33] repositories. Table 3 details the proportion of each dataset used in our pre-training mixture. Notably, beyond adhering to standard data quality filtering practices established by prior VLA works [21], we additionally ensure that all robot state annotations are accurate and physically consistent. Crucially, to empower the reasoning process without introducing computational bottlenecks during training, we precompute the DINOv3-based latent tokens for all pre-training frames offline. As detailed in Section 2.2, this involves extracting the <CLS> token from DINOv3 and applying top- k selection to generate dense, semantically rich image representations. Because these spatial and temporal latent targets are processed entirely offline, they are simply loaded alongside the standard visual and textual inputs during this pretraining stage. This strategy provides the model with highly informative cognitive anchors to learn environmental dynamics and spatial reasoning, while adding virtually zero computational overhead to the pre-training pipeline. Ultimately, this stage establishes a shared representation space across diverse robotic datasets, enabling a seamless integration of reasoning and execution within the unified VLA framework.

C.2 Real-World Data Collection

We configure the dual-arm experimental setup with two Franka Research 3 robotic arms equipped with 3D-printed UMI grippers. A single Intel RealSense D455 camera provides a third-person view, while two D435 cameras are mounted for wrist views. Inference is performed on an NVIDIA RTX

Table 3: **Datasets for Pre-training.** The names of selected datasets for large-scale pretraining and their sampling ratios (%).

Training Dataset Mixture			
Dataset	Ratio (%)	Dataset	Ratio (%)
BridgeV2 [61, 62]	20.82	Nyu Franka Play [63]	0.24
Kuka [64]	20.22	Stanford Hydra [65]	0.20
Fractal [66]	13.67	RoboMIND [33]	0.20
Robo-Net [67]	11.53	Jaco Play [68]	0.19
Language Table [69]	7.72	Dobb-E [70]	0.18
BC-Z [71]	7.54	Toto [72]	0.17
Maniskill [73]	5.26	Furniture Bench [74]	0.09
DROID [34]	4.82	Utokyo Pr2 Tabletop [75]	0.04
Roboset [76]	3.21	Utokyo Xarm Pap [77]	0.04
FMB Dataset [78]	1.50	CMU Stretch [79]	0.02
Taco Play [80, 81]	1.26	DLR Sara Grid Clamp [82]	0.02
RoboTurk [83]	0.70	Utokyo Pr2 Fridge [75]	0.01
Berkeley Autolab Ur5 [84]	0.35	—	—

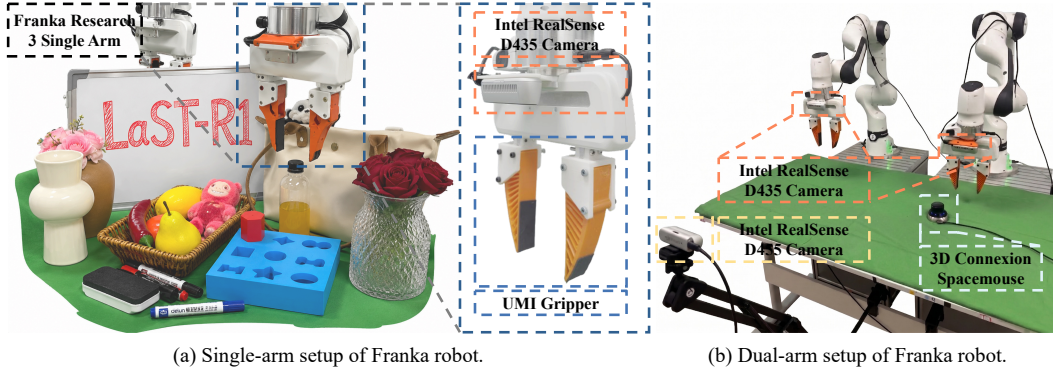


Figure 6: **Real-World Robot Experiment Setups of Franka Panda Arms.**

4090 GPU. Both SFT data collection and human-in-the-loop real-world RL are conducted using a 3D SpaceMouse. The main assets used throughout our experiments are shown in Figure 6.

1. *Insert hexagon block.* This task requires the robot to hold a hexagonal block and accurately insert it into a matching slot on a base. Success depends on the model’s ability to perform precise 6-DoF spatial alignment and maintain accurate trajectory control to handle the tight physical clearance between the object and the target.
2. *Open bag zipper.* The robot is required to use dual arms cooperatively, with one arm stabilizing a soft bag while the other grasps and pulls the zipper. This task evaluates the model’s capacity for bimanual coordination and its ability to execute a smooth, continuous pulling trajectory while interacting with a deformable object.
3. *Wipe vase with sponge.* This task requires the robot to hold a vase securely with one arm while using a sponge in the other to wipe its exterior. This demands highly accurate bimanual synchronization and the ability to predict and maintain continuous, stable contact along a curved 3D spatial trajectory.
4. *Open bottle cap.* The robot must establish a stable grasp on a bottle with one arm while the other arm twists off its cap. This task evaluates the model’s ability to execute fine motor control, specifically coordinating precise rotational manipulation and applying appropriate torque while simultaneously stabilizing the base object.

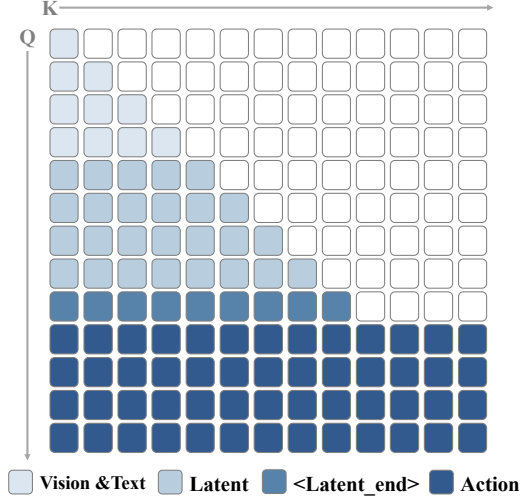


Figure 7: **Hybrid Attention Mask Design.** Our model employs a custom attention mask to unify autoregressive reasoning and parallel action execution. The vision and text prompts, alongside the latent reasoning tokens, utilize a causal lower-triangular mask for sequential generation. After the <Latent_end> transition token aggregates the full reasoning context, the action tokens employ a bidirectional mask. This allows all action tokens within a chunk to attend to the entire historical context as well as to each other, enabling efficient parallel decoding.

D Training Details

D.1 Alternative Latent Implementation Details

To systematically evaluate the architectural design of our latent reasoning tokens, we compare our proposed DINOv3-based representation against three alternative compression strategies: **1) Global Pooling**, **2) Convolutional Downsampling**, and **3) a Lightweight Q-Former** [45]. The most straightforward baseline, Global Pooling, trivially collapses the N_v visual tokens from the VLA vision encoder into a single vector via token-dimensional average pooling, which operates without additional parameters but severely sacrifices fine-grained spatial details. To explicitly model the spatial hierarchy, the Convolutional baseline reshapes the N_v visual tokens back into a 2D feature map and applies a sequence of 1×1 projections, aggressive strided downsampling (stride of 8), and global spatial convolutions with GELU activations to squeeze the map into a single flattened token. Alternatively, the Q-Former baseline employs an attention-based mechanism where a single learnable query adaptively aggregates task-relevant context from the dense N_v visual tokens through stacked Pre-LayerNorm cross-attention and MLP blocks. While these baselines introduce varying degrees of parametric complexity to compress visual features, our proposed approach simply extracts the <CLS> token from a pre-trained DINOv3 model and applies top- k dimension selection entirely offline. By directly harnessing the structurally rich and semantically dense feature space of a powerful vision foundation model, our DINOv3-based method circumvents the information loss inherent to naive pooling, spatial downsampling, or from-scratch query optimization. Consequently, this offline extraction strategy provides an optimal, computationally free cognitive anchor, ultimately achieving the best reasoning performance and highest success rates in our empirical evaluations.

D.2 Warm-up Details

Our policy is initialized with pre-trained Qwen3-VL-4B [2] weights, expanding the tokenizer vocabulary to include discrete action tokens (<action_ i > for $i \in [0, 255]$) and a special transition token (<latent_end>). As illustrated in Figure 7, we adopt a hybrid decoding architecture: latent tokens are generated autoregressively to exploit the VLM’s robust reasoning, while action tokens are decoded in parallel with a chunk size of 8 for physical execution efficiency. During training, we tailor the latent length strategy to the target domain. For the multi-task LIBERO benchmark, we encourage length exploration by uniformly sampling the reasoning length from $\{2, 4, 6, 8\}$ per sample (anchored

on a maximum length of 8 with $M = 4$ candidate positions); conversely, for single-task real-world deployments, we strictly enforce a fixed latent length of 8. The network is optimized using a joint objective comprising a cosine similarity loss for the predicted latents against offline ground-truth (GT) latents, a Cross-Entropy (CE) loss for the `<latent_end>` token, and a standard CE loss for the parallel-decoded action tokens. These components are empirically weighted at 1 : 0.1 : 1 to balance token-level gradients. Training is distributed across $8 \times \text{H20}$ GPUs using Accelerate [85] and DeepSpeed in bf16 mixed precision, with a global batch size of 64 and the AdamW [86] optimizer (peak learning rate 1×10^{-5} , cosine decay with a minimum ratio of 0.1). Finally, we customize the training duration per environment: the four LIBERO suites are trained in an extremely low-data regime (1 expert trajectory per task) for 10K iterations each, whereas real-world models are fine-tuned on 30 trajectories per task for 1K iterations.

D.3 RL Training Details on LIBERO.

We implement our Latent-to-Action Policy Optimization (LAPO) framework using ver1 [87] with Ray [88] and Fully Sharded Data Parallel (FSDP) on a single $8 \times \text{H20}$ GPU node. The VLA policy is initialized from the SFT warm-up checkpoint for continuous online interaction. During rollouts, the policy decodes an action chunk of 8 future action steps in parallel (yielding exactly 56 tokens per chunk) with a sampling temperature of 1.6, while trajectory lengths are capped at 240 steps for LIBERO-Spatial, 320 for LIBERO-Object/LIBERO-Goal, and 576 for LIBERO-Long. A strict verifier mechanism yields a sparse binary reward (0 or 1) for task success, which is scaled by a factor of 5 at the terminal step. Advantages are computed via GAE ($\gamma = 0.99$, $\lambda = 0.95$) with valid-timestep masking applied to accommodate variable-length trajectories. At each RL iteration, a rollout batch of 512 sampled trajectories is evenly partitioned into 4 mini-batches to perform 4 PPO optimization epochs. We train the policy using learning rates of 3×10^{-5} for the VLA actor and 3×10^{-4} for the value head. To ensure stability and efficiency, we apply global gradient clipping at 10, asymmetric PPO clipping bounds ($\epsilon_{\min} = 0.2$, $\epsilon_{\max} = 0.28$), and gradient/optimizer state offloading, alongside evaluating the model’s success rate every 5 update steps.

D.4 RL Training Details on Real-World

Building upon the continuous asynchronous actor-learner pipeline [89, 90], policy rollout and model optimization run concurrently without pausing data collection. The actor continuously executes the policy—which predicts actions conditioned on language prompts and dual-view RGB observations—and streams transition tuples to the learner. To preserve critical corrective signals, human interventions are routed to a dedicated buffer, allowing the learner to sample mixed mini-batches from both demonstration and rollout streams. We initialize the LaST-R1 model using the supervised warm-up checkpoint, and to ensure stable and computationally tractable online updates, we freeze the base model and exclusively update newly injected Low-Rank Adaptation (LoRA) [46] parameters with rank $r = 32$. Online optimization commences after collecting an initial replay threshold of 500 transitions. The learner optimizes a joint objective combining behavior cloning ($\lambda_{\text{BC}} = 1.0$) and Q-guided policy improvement ($\lambda_Q = 0.5$) using the AdamW [86] optimizer with a learning rate of 1×10^{-5} , zero weight decay, and gradient clipping at a maximum norm of 1.0. We apply a critic-to-actor update ratio of 2:1, where the critic utilizes one-step temporal-difference targets with a discount factor of $\gamma = 0.98$ and target value parameters are softly updated at a rate of $\tau = 0.005$. Gradient accumulation is set to 16 micro-steps with a micro-batch size of 2 per stream, yielding an effective batch size of approximately 32. Finally, to handle sparse real-world feedback, the environment provides a terminal reward of +10 upon operator-confirmed task completion, coupled with a constant step penalty of -0.05 to encourage execution efficiency, while the learner periodically broadcasts updated weights back to the actor for hot-loading.

E Additional Quantitative Analysis

E.1 Additional Ablation Studies on Hyperparameters

To systematically evaluate the impact of the weighting hyperparameters in our joint training objective $\mathcal{L}_{\text{total}}(\theta)$, we conduct an ablation study varying the coefficients λ_1 , λ_2 , and λ_3 on LIBERO-Spatial. As shown in Figure 8, in each experiment, we isolate the effect of one coefficient while fixing the

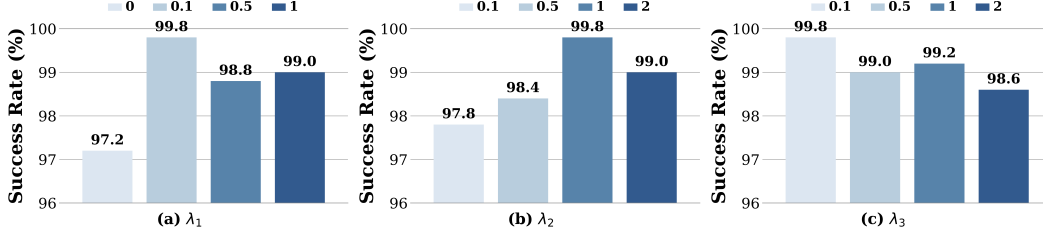


Figure 8: **Ablation Studies on Loss Coefficients.** Performance impact of varying (a) latent loss weight λ_1 , (b) state-value weight λ_2 , and (c) transition penalty λ_3 . In each sub-figure, unlisted coefficients are fixed at their optimal values. The combination ($\lambda_1 = 0.1, \lambda_2 = 1, \lambda_3 = 0.1$) achieves the optimal gradient balance and highest success rate.

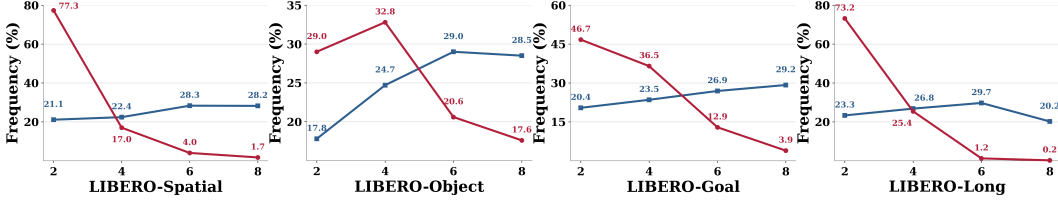


Figure 9: **Frequency Distribution of Adaptive Latent Reasoning Lengths.** Compared to the SFT warm-up (blue), the RL-optimized policy (red) successfully learns the optimal number of latent reasoning steps, achieving an excellent balance between reasoning precision and inference efficiency.

others at their empirically optimal values (i.e., $\lambda_1 = 0.1, \lambda_2 = 1, \lambda_3 = 0.1$). First, analyzing the latent loss weight λ_1 , we observe that relying entirely on implicit gradient backpropagation from the action loss ($\lambda_1 = 0$) yields a suboptimal success rate of 97.2%. Introducing explicit latent supervision significantly improves performance, peaking at 99.8% with $\lambda_1 = 0.1$. However, excessively high values (e.g., $\lambda_1 = 1$) slightly degrade performance to 99.0%, likely because the latent objective begins to overshadow the primary physical execution task. Second, for the state-value estimation weight λ_2 , a balanced weight of $\lambda_2 = 1$ achieves the best result (99.8%). Lower weights such as 0.1 and 0.5 degrade the success rate to 97.8% and 98.4% respectively, underscoring the necessity of robust value estimation for accurate advantage computation in our LAPO framework. Finally, evaluating the transition weight λ_3 reveals that a modest penalty of $\lambda_3 = 0.1$ is optimal (99.8%). Increasing this weight up to 2 forces a sharp performance drop to 98.6%, indicating that over-penalizing the `<latent_end>` token disrupts the delicate exploration balance required for the policy to learn state-conditional reasoning lengths. Consequently, the configuration of $\lambda_1 = 0.1, \lambda_2 = 1$, and $\lambda_3 = 0.1$ establishes the effective balance across action generation, internal reasoning, and value estimation.

E.2 Analysis of Adaptive Reasoning Length

To investigate the efficacy of our dynamic reasoning mechanism, we analyze the frequency distribution of the chosen latent reasoning lengths (bounded by a maximum of 8 tokens) across the four LIBERO task suites. Figure 9 illustrates the contrast between the policy right after SFT warm-up and the fully RL-optimized policy. During the warm-up phase, where reasoning lengths are sampled randomly, the pre-RL model exhibits a relatively uniform distribution across lengths 2, 4, 6, and 8. This indicates a lack of cognitive prioritization, treating simple and complex states with equal computational budgets. Following Latent-to-Action Policy Optimization (LAPO), the length distribution undergoes a dramatic shift. Guided by the environment reward and our transition-specific optimization objective, the model effectively learns an "early-exit" strategy. Across all environments, the RL-optimized policy heavily gravitates toward shorter cognitive horizons, with reasoning lengths of 2 or 4 tokens dominating the decision steps. This empirical behavior strongly validates our design: rather than passively executing fixed-length reasoning, LaST-R1 proactively learns to terminate its internal deliberation once sufficient state representations are formed. By drastically reducing the token count per step without compromising the near-perfect success rates, the model achieves an optimal balance between robust physical reasoning and inference efficiency.

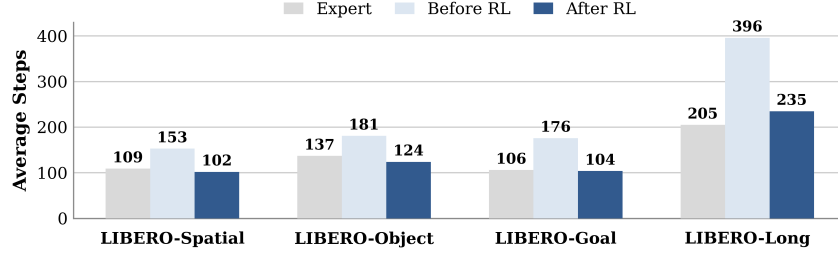


Figure 10: **Comparison of Average Execution Steps Across LIBERO Task Suites.** We report the average steps taken by the expert demonstrations, the pre-RL policy, and the RL-optimized policy. After optimized via LAPO, LaST-R1 not only drastically reduces the execution steps compared to the imitation baseline, but even surpasses the temporal efficiency of experts in most scenarios.

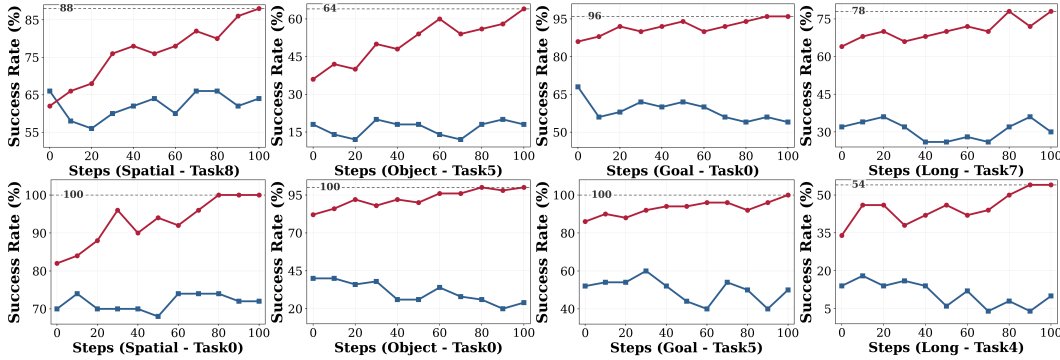


Figure 11: **Generalization Analysis on LIBERO.** For each task suite, models are warmed up with one trajectory per task, followed by online RL training on 9 tasks, with the remaining 1 task for evaluation. While the out-of-distribution performance of the Action-Only PPO baseline (blue) stagnates, our LaST-R1 with LAPO (red) shows continuous generalization improvements across all task suites.

E.3 Episode Length Comparisons

To further demonstrate the efficacy of our framework, we evaluate the average number of execution steps across 500 test trajectories for each LIBERO suite. Notably, these averages are computed unconditionally over all rollouts, regardless of ultimate task success or failure. As illustrated in Figure 10, the policy trained after warm-up SFT (Before RL) requires significantly more steps than the simulator-provided expert demonstrations. This inflation in trajectory length stems from compounding errors and purely reactive behaviors; when the pre-RL model encounters out-of-distribution (OOD) states, it often gets trapped in inefficient loops or prolonged, unsuccessful wandering that approaches the maximum step limit. However, following online interaction, the RL-optimized policy (After RL) exhibits a dramatic reduction in execution steps, notably outperforming even the original expert trajectories in the Spatial, Object, and Goal task suites. We attribute this exceptional temporal efficiency directly to the optimized latent reasoning process. Rather than passively mimicking the scripted expert data, which often relies on conservative motion planning and rigid, waypoint-based trajectories, the RL-driven reasoning strategy empowers the policy with a deeper, more generalized understanding of the task dynamics. By jointly optimizing the reasoning and action spaces, the model learns to synthesize highly efficient internal representations, thereby discovering optimal shortcuts and formulating more direct, precise, and decisive trajectories than the original algorithmic experts.

E.4 Additional Generalization Analysis

To provide a more granular understanding of the out-of-distribution (OOD) learning process, Figure 11 details the step-by-step generalization curves for two representative held-out tasks from each LIBERO suite. The learning dynamics reveal a stark contrast between the two paradigms throughout the online interaction phase. The Action-Only PPO baseline (blue) exhibits a classic overfitting

pathology. Across almost all tasks, its OOD success rate flatlines early in training and frequently oscillates or even collapses (e.g., Spatial-Task8 and Long-Task7). This empirical evidence confirms that optimizing purely in the action space forces the model to memorize the specific kinematic trajectories of the 9 training tasks, leaving it entirely brittle and incapable of adapting when faced with the unseen spatial configurations or objects of the held-out task. In sharp contrast, our LaST-R1 optimized with LAPO (red) displays a remarkably stable and continuous upward trajectory across the entire training process. The curves demonstrate no signs of catastrophic forgetting or OOD performance degradation. Notably, our policy not only rapidly converges to a perfect 100% OOD success rate on several tasks (Spatial-Task0, Object-Task0, and Goal-Task5), but also shows robust, monotonic growth on the most challenging multi-stage tasks. For instance, in Long-Task4, while the action-only baseline completely fails to surpass a 20% success rate due to compounding errors, our method steadily climbs to 54%. This extended analysis clearly illustrates that jointly optimizing the latent reasoning space empowers the policy to abstract high-level semantic and physical principles, allowing it to dynamically compose learned skills for novel scenarios rather than rigidly overfitting to the training distribution.

F Additional Qualitative Analysis

F.1 Action-to-Vision Attention

To gain deeper insights into the internal decision-making mechanisms of our policy, we utilize Grad-CAM [91] to visualize the cross-attention weights from the action tokens to the visual tokens. As illustrated in Figure 16, we compare spatial attention maps across four representative trajectories from the LIBERO task suites to evaluate the impact of explicit latent reasoning and our LAPO framework. Examining the SFT-trained models, the Action-Only policy frequently exhibits diffused and scattered attention, struggling to consistently localize task-relevant entities and often lagging behind the robot’s end-effector. In stark contrast, by generating latent reasoning tokens prior to action decoding, our proposed LaST-R1 establishes a strong semantic anchor, resulting in highly concentrated and precise visual grounding on critical objects of interaction. These advantages become even more pronounced after reinforcement learning post-training. While applying standard PPO to the Action-Only model tends to over-focus on the gripper’s immediate vicinity and lacks long-horizon awareness, LaST-R1 + LAPO achieves the most robust, goal-directed attention. Because LAPO jointly optimizes the latent reasoning space and the action space using the environment reward, the policy learns to dynamically shift its intense visual focus from the manipulated object to the target receptacle as the trajectory progresses, confirming that our framework successfully aligns internal visual cognition with precise physical execution.

F.2 Real-World Visualization

To evaluate the practical deployment capabilities of LaST-R1, we visualize its execution trajectories across diverse real-world manipulation tasks, as shown in Figure 12. As demonstrated, the policy exhibits stable closed-loop control during standard execution (e.g., precise insertion, bimanual coordination). Crucially, the model demonstrates remarkable robustness in visually cluttered environments, as shown in Figure 13. Despite severe out-of-distribution perturbations—such as dynamic lighting, novel distractor objects, and background variations—LaST-R1 consistently maintains execution stability, highlighting its strong generalization capabilities in complex, unconstrained scenarios.

G Failure Case Analysis

G.1 Simulation.

In the rare failure cases observed, the model’s errors stem from fine-grained physical execution. As shown in Figure 14, in the LIBERO-Spatial task: *pick up the black bowl next to the plate and place it on the plate*, the robot successfully grasps the bowl but releases it slightly off-center, causing it to slip off the edge of the plate due to imperfect placement precision. Similarly, in the LIBERO-Long task: *turn on the stove and put the moka pot on it*, while the robot correctly turns on the stove, its arm inadvertently knocks over the moka pot during the approach phase. This irreversible change

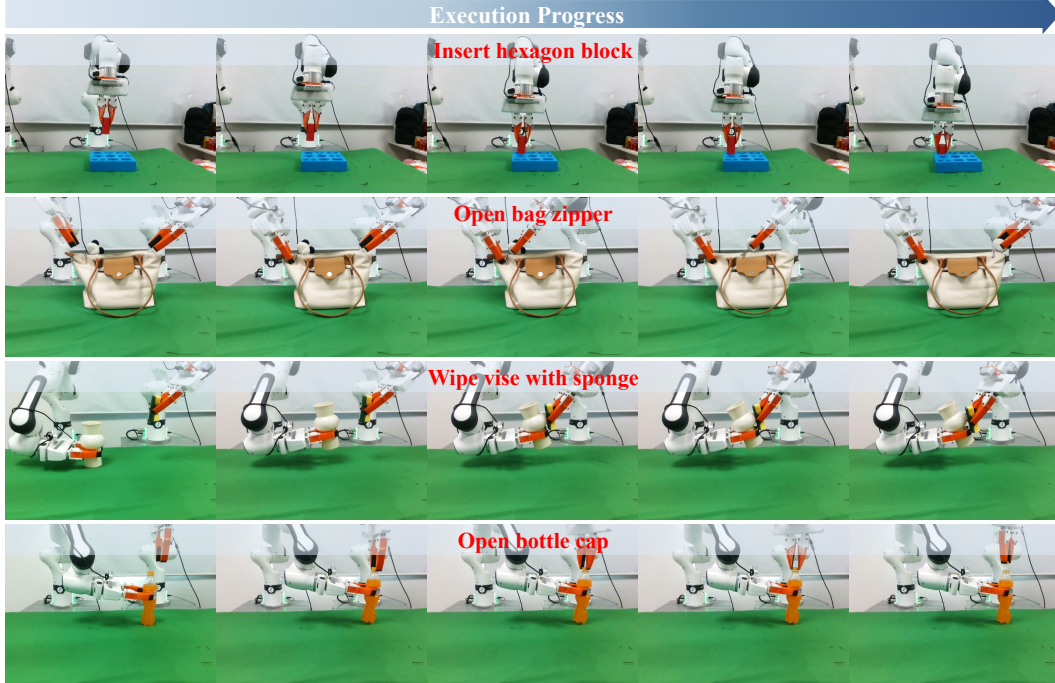


Figure 12: **Real-World Execution Trajectories of the Proposed Policy.** The sequences illustrate the continuous execution progress across diverse robotic manipulation tasks.

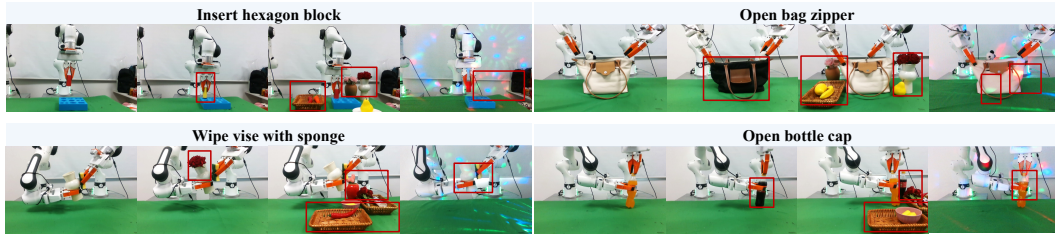


Figure 13: **Policy Robustness Under Unseen Configurations.** The model consistently maintains task success when subjected to severe visual variations, effectively ignoring out-of-distribution distractors, unseen object instances, and dynamic lighting conditions (indicated by red bounding boxes).

in the object’s pose prevents subsequent grasping, highlighting the inherent long-tail challenge of maintaining strictly collision-free trajectory planning in complex, multi-step environments.

G.2 Real-World.

In real-world deployments, the observed failure cases primarily stem from fine-grained execution imperfections and complex contact dynamics. As shown in Figure 15, for the *Insert hexagon block* task, the failure is caused by a slight positional deviation during the final insertion phase, preventing the block from fitting into the slot. In the *Open bag zipper* task, the robot struggles with dynamic friction, causing the gripper to slip and detach from the zipper mid-pull. For the *Wipe vase with sponge* task, the failure occurs because the robot fails to establish actual physical contact between the sponge and the vase surface, resulting in an ineffective wiping motion. Finally, in the *Open bottle cap* task, the robot successfully grasps and twists the cap but terminates the action prematurely, failing to unscrew the cap completely. These cases indicate that although LaST-R1 achieves remarkable performance in handling continuous, high-precision physical feedback in real-world contact-rich scenarios, a certain probability of failure still exists.

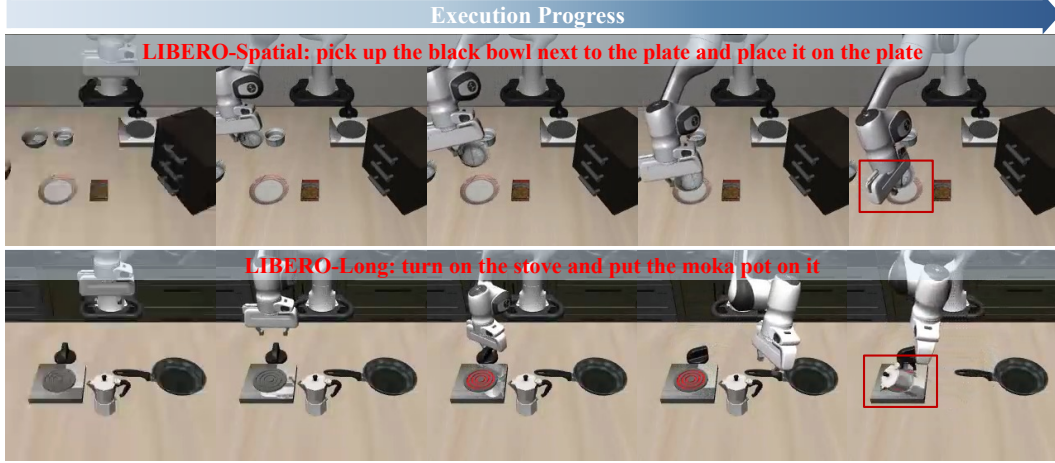


Figure 14: **Failure Cases in the LIBERO.** Red boxes highlight the moments of failure.

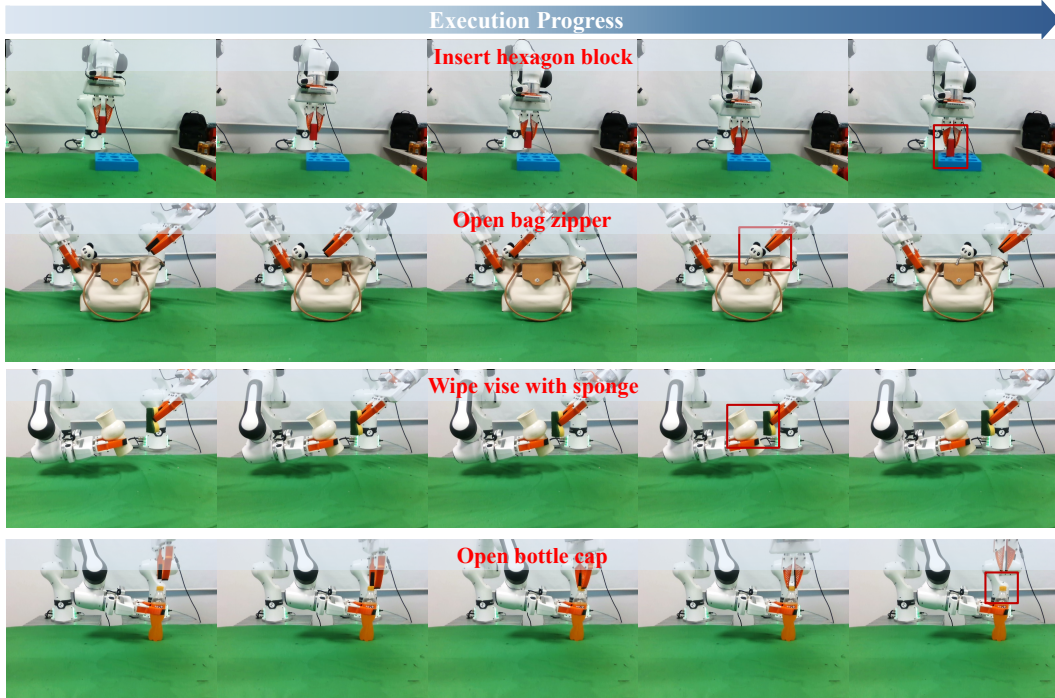


Figure 15: **Visualizations of Real-World Failure Cases.** The sequences illustrate typical execution errors across four tasks, with red bounding boxes highlighting the specific moments of failure.

H Limitations and Future Work

While LaST-R1, optimized via the proposed LAPO framework, exhibits exceptionally strong performance across both the LIBERO benchmark and real-world deployments, it has several limitations that present exciting avenues for future research. First, to strike an optimal balance between reasoning capacity and inference speed, our current adaptive CoT mechanism caps the maximum latent sequence length at 8 and restricts the early-exit transitions to a predefined set of M candidate positions. Moving forward, we aim to refine the latent representation mechanism and explore more flexible, fully dynamic reasoning architectures that can halt deliberation without manually designed constraints. Second, our current real-world evaluations are conducted exclusively on single-arm and dual-arm manipulation platforms. To demonstrate the scalability and universality of our framework, we plan to

deploy LaST-R1 to more complex robotic embodiments, extending our latent reasoning paradigm to multi-fingered dexterous hands and whole-body humanoid robot control.

I Broader Impact

Our work proposes LaST-R1 VLA, a foundation model for robotic manipulation that seamlessly integrates adaptive-length latent reasoning and parallel action execution within a unified framework. While our method significantly enhances spatial awareness and dynamically allocates cognitive compute through Latent-to-Action Policy Optimization (LAPO), deploying high-capacity VLA models in physical environments inherently introduces potential risks. These risks primarily involve unpredictable physical behaviors under out-of-distribution scenarios or misinterpretation of ambiguous human instructions. To mitigate these challenges, future real-world deployments should incorporate rigorous safety guardrails and strict hardware-level operational constraints. Broadly, our post-training framework delivers strong performance across diverse downstream tasks, facilitating the transition of robotic systems from controlled laboratory settings to real-world deployment. Meanwhile, by enabling robots to fluidly balance deliberate cognitive planning with rapid physical reflexes, our approach advances the development of reliable, general-purpose robotic assistants for complex, unstructured domains such as smart manufacturing, elder care, and domestic automation.

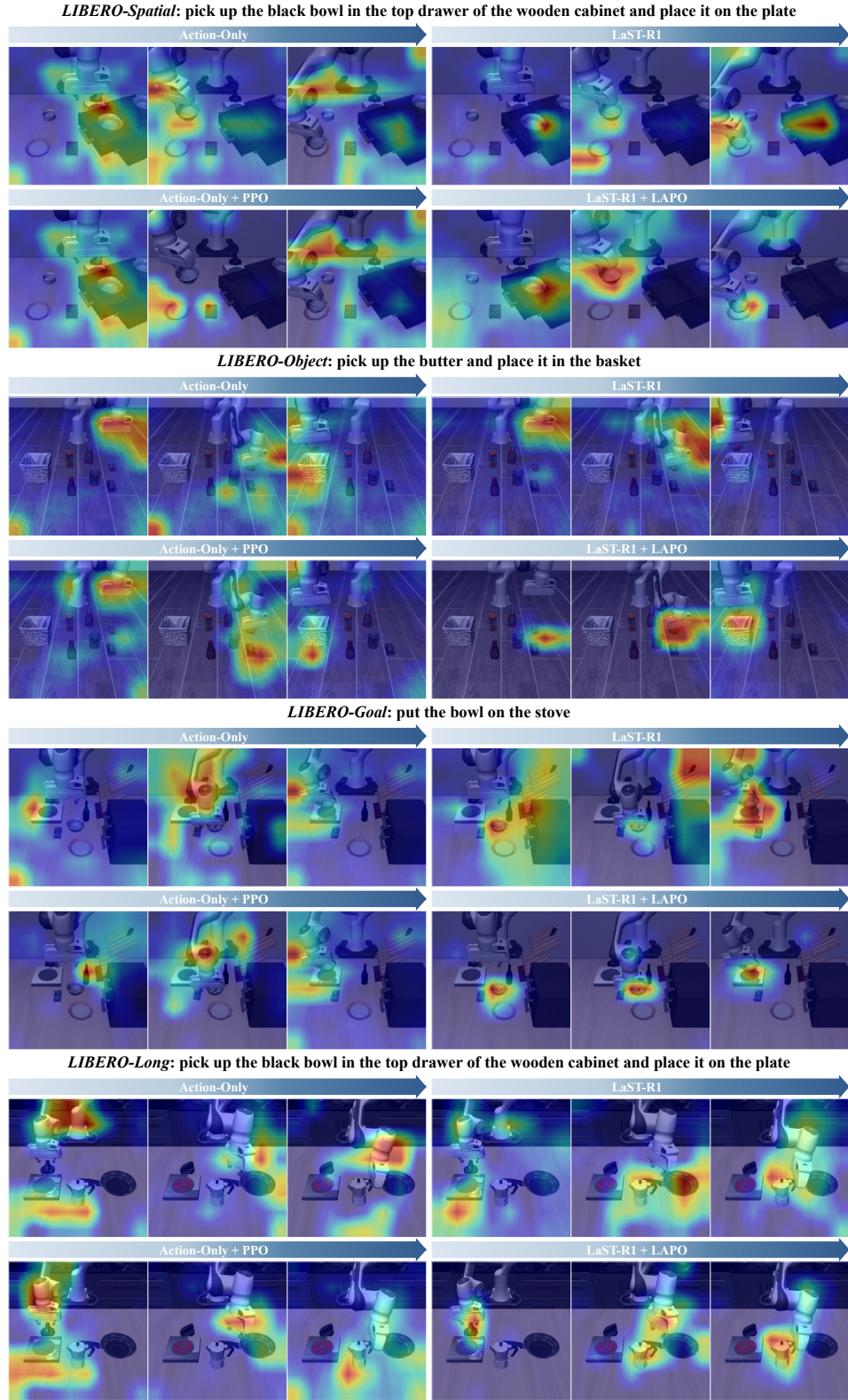


Figure 16: **Visualizations of Action-to-Vision Cross-Attention.** We compare the attention maps across four representative trajectories from four task suites in LIBERO.